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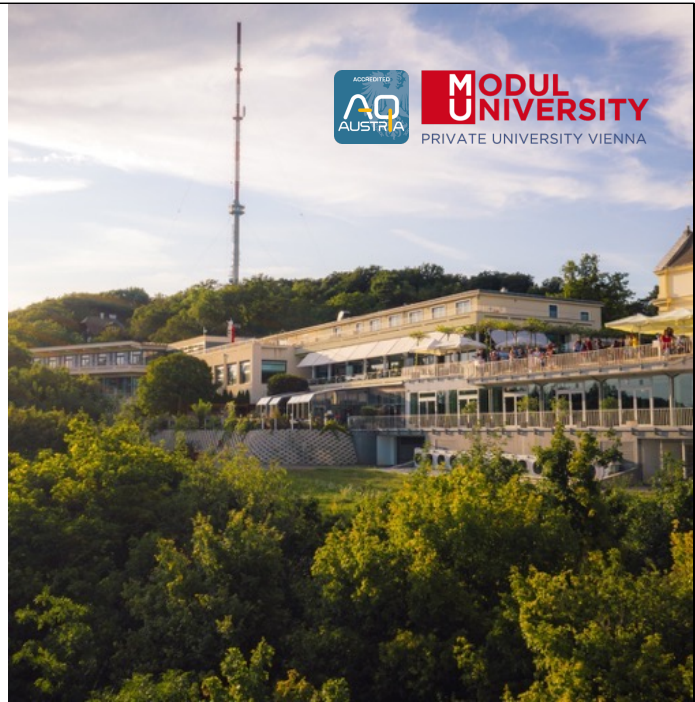
SEID 2364

SESSION 12

Ethical Systems as Strategic Games

David B. Smith

*Agents, Incentives,
Signals, and
Responsibility*



Establish continuity from the prior weeks. This session does not introduce ethics from scratch. Students have already worked with ethical theory, the EU AI Act, mediation pathways, risk, distance, authority, trust, and EDOCA. Today extends that work by asking how ethical responsibility changes when systems include uncertainty, misinformation, bad actors, professional obligations, and conflicting frameworks.

Today's Activities

GitHub and the assignment status.

Game as fruitful approach to analyzing ethics

Game Theory as an analytical tool

INSTRUCTOR NOTES:

Set the intellectual arc. Students are not abandoning ethical frameworks, BBS, EDOCA, or professional codes. They are adding a strategic modeling layer that helps organize case analysis.

This is a synthesis session, not a new disconnected topic.

SPEAKING NOTES:

"We are not replacing the tools we have used. We are adding a structure that helps us see how agents, goals, rules, information, and signals interact."

A Game Is a System, Not Play

In game theory, a **game** is any formal mathematical model of strategic interaction.

It defines a situation where **multiple decision-makers interact**, and the ultimate outcome — or **payoff** — for each participant depends not only on their own choices, but also on the choices made by others.

A game is specified by:

- **Players** — the decision-makers
- **Strategies** — the choices available to each player
- **Payoffs** — the outcomes determined by all players' choices together
- **Information** — what each player knows at the point of decision
- **Rules** — the constraints on allowable actions

INSTRUCTOR NOTES:

Establish the formal game theory definition immediately. A game is any formal mathematical model of strategic interaction. It is not play or amusement. It is a system that describes a situation where multiple decision-makers interact and each participant's payoff depends not only on their own choices but on the choices made by others. Walk through the five components: players, strategies, payoffs, information, and rules.

Emphasize that the defining feature is strategic interdependence: each player's outcome is a function of what all players do, not just their own choices. This is what separates a game from an individual optimization problem and what makes it a system rather than a collection of independent decisions.

SPEAKING NOTES:

"In game theory, a game is a formal mathematical model. It describes a situation where multiple decision-makers interact, and the payoff for each depends not just on what they choose, but on what everyone else chooses. That mutual dependence is what makes it a system. The components are players, strategies, payoffs, information, and rules."

What Game Theory Means by "Game"

A game is a formal model of strategic interaction.

A game includes:

- agents
- actions
- rules
- information
- incentives
- outcomes

INSTRUCTOR NOTES:

Clarify the term "game." The word does not mean "play, amusement, or triviality. It means a structured decision environment.

This is the first core definition.

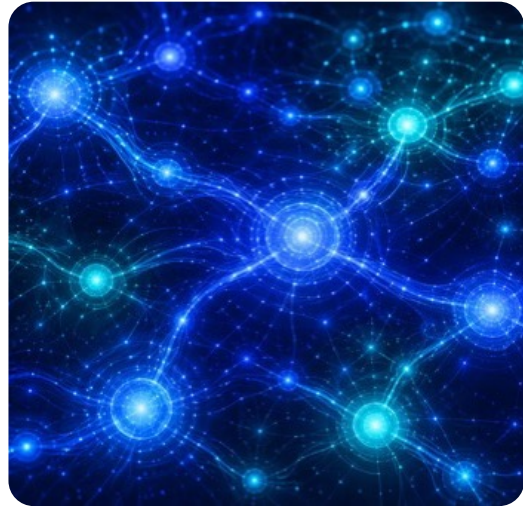
SPEAKING NOTES:

"In game theory, a game is a simplified formal model. It lets us describe who can act, what they know, what they want, and what happens when they act."

Strategic Interdependence

Strategic interdependence:

One agent's action changes the conditions, options, risks, or outcomes available to another agent.



INSTRUCTOR NOTES:

This is the key term for the mini-lecture.

Make sure students understand that a system becomes ethically and strategically complex when agents affect one another through the structure of the system.

SPEAKING NOTES:

"If a user posts, a platform ranks, a regulator reacts, and a bad actor adapts, none of those actions are independent."

The Payoff Matrix

A **payoff matrix** is a table that displays the outcome for every possible combination of player strategies.

It makes a game visible by showing:

- **Players** — one on each axis
- **Strategies** — each row and column is a possible choice
- **Payoffs** — each cell shows the outcome for both players

	Player B: Cooperate	Player B: Defect
Player A: Cooperate	(3,3)	(0,5)
Player A: Defect	(5,0)	(1,1)

Can the structure make the best collective outcome individually irrational?

INSTRUCTOR NOTES:

Introduce the payoff matrix as the fundamental representation tool in game theory. It is not specific to any particular game — it is the general method for displaying strategic interaction. Rows represent one player's strategies, columns represent the other's, and each cell contains the payoff pair. Emphasize that the matrix makes interdependence concrete: you cannot read your payoff without also reading the other player's choice.

The closing question — whether the structure can make the best collective outcome individually irrational — sets up the prisoner's dilemma on the next slide.

SPEAKING NOTES:

"A payoff matrix is the basic tool for making a game visible. Players on the axes, strategies on the rows and columns, outcomes in the cells. The key insight is that you cannot determine your outcome without knowing what the other player chose. That is strategic interdependence in one table."

Rock Paper Scissors

Payoffs are (Player 1, Player 2)

		PLAYER 2		
		 Rock	 Paper	 Scissors
PLAYER 1	 Rock	$(0, 0)$ (Draw)	$(-1, 1)$ (Player 2 wins)	$(1, -1)$ (Player 1 wins)
	 Paper	$(1, -1)$ (Player 1 wins)	$(0, 0)$ (Draw)	$(-1, 1)$ (Player 2 wins)
	 Scissors	$(-1, 1)$ (Player 2 wins)	$(1, -1)$ (Player 1 wins)	$(0, 0)$ (Draw)

The Prisoner's Dilemma

Two players each choose to **cooperate** or **defect**, without knowing the other's choice.

- Both cooperate — good outcome for both (3, 3)
- One defects, one cooperates — defector wins big, cooperator loses (5, 0)
- Both defect — poor outcome for both (1, 1)

The dilemma: Regardless of what the other player does, each player is individually better off defecting. Both defect — even though both cooperating would leave everyone better off.

Individual rationality produces collective harm. This is the core tension game theory makes visible.

INSTRUCTOR NOTES:

Present the prisoner's dilemma as the canonical example of a payoff matrix in action. This is the single most important game in game theory and the structural foundation for everything that follows in this session.

Walk through the dominant strategy logic step by step:

1. Assume the other player cooperates. You get 3 by cooperating or 5 by defecting. Defecting is better.
2. Now assume the other player defects. You get 0 by cooperating or 1 by defecting. Defecting is still better.
3. Conclusion: defection is better regardless of what the other player does. This is a dominant strategy.
4. Both players follow identical logic. Both defect. The result is (1, 1) — the Nash Equilibrium.
5. But (3, 3) — mutual cooperation — is better for both. The equilibrium is Pareto-inferior.

Key terms to introduce here: dominant strategy, Nash Equilibrium, Pareto optimality. Students do not need to memorize formal definitions, but they need

to understand the intuition: a Nash Equilibrium is an outcome where no player can do better by unilaterally changing strategy; a Pareto-optimal outcome is one where no player can be made better off without making another worse off. The prisoner's dilemma is powerful precisely because the equilibrium is not Pareto-optimal.

This is the foundational structure students need before encountering the platform moderation game on the next slides. The prisoner's dilemma demonstrates that individual rationality can produce collective harm — a pattern that recurs throughout data ethics, regulatory design, and sociotechnical systems.

Anticipated student question: "Why can't they just agree to cooperate?" This is the right question. The answer depends on enforcement, repetition, trust, and institutional design — which is exactly where governance and ethics enter the analysis. Do not resolve this yet; let it motivate the next slides.

SPEAKING NOTES:

"This is the prisoner's dilemma — the most famous game in game theory. Let me walk through the logic. Suppose the other player cooperates. You get 3 by cooperating, or 5 by defecting. Defecting is better. Now suppose the other player defects. You get 0 by cooperating, or 1 by defecting. Defecting is still better. No matter what the other player does, you are better off defecting. Both players reason this way. Both defect. Both end up at (1, 1) — even though (3, 3) was available if they had both cooperated."

"The structure of the game punishes cooperation and rewards defection. Individual rationality produces collective harm. That tension is the dilemma — and it is the same tension we will see in platform moderation, data sharing, and regulatory compliance."

Payoff Matrix: Before Governance

Shareholder perspective. Two platforms each choose: responsible moderation or lax/engagement-first. No governance mechanisms are in play.

	Platform B: Moderate	Platform B: Lax
Platform A: Moderate	Both retain users, share high costs, strong reputation <i>(Best collective outcome — but costly for both)</i>	A loses users and revenue while bearing moderation costs alone <i>(A is exploited — responsible but uncompetitive)</i>
Platform A: Lax	A gains users and revenue at low cost while B bears moderation burden <i>(B is exploited — A free-rides on B's investment)</i>	Neither invests in moderation. No competitive edge for either. Ecosystem degrades for everyone. <i>(Nash Equilibrium — stable but collectively the worst outcome)</i>

The logic: Regardless of what B does, A is always better off choosing lax. The same logic applies to B. Both end up lax — even though both moderating would leave everyone better off.

INSTRUCTOR NOTES:

Explain the numbers as simplified strategic payoffs. They combine engagement, cost, reputation, regulatory exposure, and ecosystem health. The numbers are not literal money.

Walk through the logic:

- If Platform B moderates, Platform A does better by being lax: 4 instead of 3.
- If Platform B is lax, Platform A still does better by being lax: 2 instead of 1.
- The same logic applies to Platform B.

SPEAKING NOTES:

"The non-trivial result is that each platform has an incentive to choose lax moderation, even though both would be better off if both moderated responsibly."

Nash Equilibrium

Definition:

A Nash Equilibrium is an outcome in a strategic game where no individual player can improve their result by unilaterally changing their own strategy, given what every other player is doing.

Key properties:

- Stability — once reached, no one has an incentive to deviate alone
- Self-enforcing — requires no external authority to maintain
- Not necessarily optimal — a stable outcome can be collectively worse than an available alternative

Why it matters for ethics:

Systems can settle into stable but harmful outcomes — not because agents are ignorant or malicious, but because the incentive structure makes harm the rational individual choice.

INSTRUCTOR NOTES

Core concept (keep this tight — 3–4 minutes max):

Define Nash Equilibrium in plain language. Do not teach the formal mathematics. The definition on the slide is sufficient: an outcome where no player can improve by changing strategy alone, given what everyone else is doing.

Three properties to emphasise:

1. Stability — once reached, no one has an incentive to deviate alone. This is what makes it “stick.”
2. Self-enforcing — requires no external authority to maintain. The structure does the work.
3. Not necessarily optimal — a stable outcome can be collectively worse than an available alternative. This is the critical ethical insight.

What the matrix shows (connect back to Before Governance slide):

The stable outcome is not always the ethical outcome. Ethical failure can emerge from:

- Incentives — each player is individually rational, but the collective result is harmful

- Costs — responsible behaviour is expensive; irresponsible behaviour is cheap
- Competition — unilateral responsibility puts you at a disadvantage
- Weak governance — no mechanism to reward cooperation or punish defection
- Asymmetric risk — the costs of being exploited fall unevenly

Point out that all five of these are visible in the Before Governance payoff matrix.

Ask students to identify which cell illustrates which factor.

Why it matters for ethics (the course-level point):

Systems can settle into stable but harmful outcomes — not because agents are ignorant or malicious, but because the incentive structure makes harm the rational individual choice. This is not a failure of character. It is a failure of system design. That distinction is central to the rest of the course.

Anticipated student questions:

“Can’t they just agree to cooperate?” — Yes, if they can make binding commitments. That’s exactly what governance does. Preview the After Governance slide.

“Is there always a Nash Equilibrium?” — In finite games with mixed strategies, yes (Nash’s theorem). But don’t go deep here. Acknowledge and move on.

“What if there are more than two players?” — Good instinct. Real systems have many players. The COVID case will have at least five. The 2x2 matrix is a teaching tool, not a ceiling.

SPEAKING NOTES

“A Nash Equilibrium is what happens when every player is doing the best they can, given what everyone else is doing. No one has a reason to change — alone. The problem is: that stable point might be terrible for everyone. That’s exactly what we saw in the Before Governance matrix. Both platforms choosing lax is stable. It’s also the worst collective outcome. The structure of the game drives the result — not bad intentions.”

“So the question becomes: if the stable outcome is harmful, what can change the game? That’s where governance comes in — and we’ll see that in the next matrix.”

The Nash Equilibrium Use in Computer Science

In Computer Science: [Nash Equilibrium](#) is used in computer science to study algorithms and decision-making. Game theory is used to develop algorithms that can make decisions in situations of uncertainty. For example, in the game of chess, each player has to consider the possible moves of the other player before making any decisions. This scenario is an example of the [Nash Equilibrium](#), where each player has to consider the actions of the other player before making any decisions.

Source <https://fastercapital.com/topics/nash-equilibrium-and-the-payoff-matrix.html/1>

Key point: This slide bridges from the abstract game theory concepts we've been building (Prisoner's Dilemma, payoff matrices) into a domain students are already familiar with — computer science.

Teaching angle: Emphasise that Nash Equilibrium isn't just an economics concept — it underpins real algorithmic design. Minimax algorithms in chess, auction mechanisms in ad tech, routing protocols in networks — all rely on reasoning about what a rational opponent or agent would do. The chess example is intuitive but be ready to extend: ask students where else they've seen “anticipate the other player's move” logic in CS (e.g., adversarial machine learning, cybersecurity, mechanism design).

Nuance to flag: Chess is technically a perfect-information game, so it's not the strongest Nash Equilibrium example. If a student raises this, acknowledge it — the deeper applications are in imperfect-information settings like poker AI (Libratus, Pluribus) where Nash Equilibrium strategies are computed to avoid exploitation.

Connection to the course theme: The ethical dimension matters here too. Algorithmic decision-making that converges on a Nash Equilibrium can produce

stable but socially harmful outcomes — the same structural problem we saw with platform moderation. This is why “ethical AI” can’t just mean “good intentions” — it requires understanding the game-theoretic incentives baked into the system.

The Prisoner's Dilemma

The classic illustration of Nash Equilibrium. Two suspects are arrested and interrogated separately. Each must choose: cooperate with the other (stay silent) or defect (betray).

	Prisoner B: Cooperate (Silent)	Prisoner B: Defect (Betray)
Prisoner A: Cooperate (Silent)	Both serve 1 year <i>(Best collective outcome)</i>	A serves 3 years, B goes free <i>(A is exploited)</i>
Prisoner A: Defect (Betray)	A goes free, B serves 3 years <i>(B is exploited)</i>	Both serve 2 years <i>(Nash Equilibrium — stable but worse)</i>

The logic: Regardless of what the other does, each prisoner is better off defecting. But when both defect, the outcome is worse for both than if both had cooperated.

The structural point: The payoff matrix defines the game. The Nash Equilibrium identifies which outcome the game settles into — here, both defect (both serve 2 years), even though both cooperating (1 year each) would be better for everyone. The matrix is the structure; the equilibrium is the result.

Nash Equilibrium: Why the Worst Outcome Persists

Recall: Two platforms each choose between responsible moderation and lax/engagement-first moderation. Payoffs are evaluated from the shareholder perspective across revenue, cost, reputation, and user trust.

The stable outcome is not the best outcome.

Each platform reasons independently:

"If the other moderates, I gain by going lax."

"If the other is lax, I still gain by going lax."

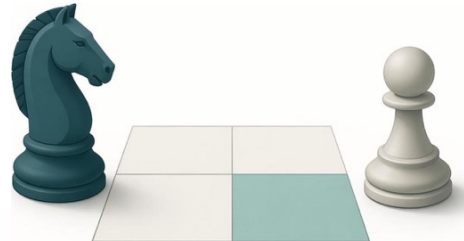
Both platforms reach the same conclusion → both choose lax.

Nash Equilibrium: Both Lax

Revenue: Moderate | Cost: Low

Reputation: Weak | User Trust: Low

The ethical insight: Rational self-interest by each platform produces a collectively worse outcome. Neither platform is ignorant or malicious — the structure of the game drives the result.



Not Every Game Has a Nash Equilibrium

Rock Paper Scissors

No pure-strategy Nash Equilibrium.

Every choice loses to one and beats another
If you pick Rock, I switch to Paper
If I pick Paper, you switch to Scissors
The cycle never settles — no stable outcome

Prisoner's Dilemma

Has a clear Nash Equilibrium: both defect.

Defection is dominant — better no matter what
Neither player gains by switching alone
The outcome is stable but collectively harmful

Why does this matter?

A Nash Equilibrium tells us where a system **settles**. If no equilibrium exists in pure strategies, behavior keeps cycling — no prediction, no stability.

When an equilibrium **does** exist, we can ask the ethical question: Is the stable outcome also the responsible one?

In the prisoner's dilemma, the answer is **no**. The system settles at the worst collective outcome. That is why governance, regulation, and institutional design matter — they change the game so the equilibrium shifts.

INSTRUCTOR NOTES:

This slide contrasts two games to sharpen understanding of Nash Equilibrium. The point is that Nash Equilibrium is not a universal feature of all games — some games have one, some do not (in pure strategies).

Rock Paper Scissors: Walk through why no pure-strategy equilibrium exists. If Player A picks Rock, Player B wants Paper. But if B picks Paper, A wants Scissors. If A picks Scissors, B wants Rock. The best-response cycle never terminates. There is no pair of pure strategies where both players are simultaneously best-responding. (Note for advanced students: RPS does have a mixed-strategy Nash Equilibrium where each player randomizes equally across all three choices. You may mention this but do not dwell on it — the pedagogical point is about the absence of a stable pure-strategy outcome.)

Prisoner's Dilemma: Recall the dominant strategy logic from the previous slide. Both defect is a Nash Equilibrium because neither player can improve by switching unilaterally. If A switches from Defect to Cooperate while B stays at Defect, A goes from 1 to 0 — worse. This is what makes the outcome stable and what makes it ethically troubling: the harmful outcome persists because no

individual actor has an incentive to change.

The right-column content connects this to the ethical stakes: when a system has a stable but harmful equilibrium, only external intervention — governance, regulation, institutional design — can shift it. This sets up the governance payoff matrix that follows.

SPEAKING NOTES:

“Not every game has a Nash Equilibrium. Rock Paper Scissors is the classic example. If I pick Rock, you switch to Paper. If you pick Paper, I switch to Scissors. The cycle never stops — no pair of choices is stable. There is no outcome where both players are satisfied with their choice given the other’s.”

“The prisoner’s dilemma is different. Both defect IS a Nash Equilibrium. Neither player can do better by changing alone. That stability is exactly what makes it dangerous — the system locks into a collectively bad outcome, and no individual has a reason to move. To break the lock, you need to change the game itself. That is what governance does.”

Ethics Changes the Game

Governance can change payoffs.

Possible mechanisms:

- professional standards
- regulation
- audits
- liability
- transparency requirements
- reputational penalties
- shared industry agreements



INSTRUCTOR NOTES:

Connect directly to ACM, IEEE, EU governance, and professional responsibility. Ethics is not only advice to individuals. Ethical and governance systems can alter the strategic environment by changing incentives, costs, rewards, observability, and authority.

SPEAKING NOTES:

"Professional ethics and regulation can function as game-changing mechanisms. They can make irresponsible behavior more costly and responsible behavior more viable."

Payoff Matrix: After Governance

Shareholder perspective. Governance mechanisms are now in play: transparency requirements, regulatory penalties for negligence, and industry standards. The incentive structure has shifted.

	Platform B: Moderate	Platform B: Lax
Platform A: Moderate	Both comply with standards, share costs, and build durable trust with users and regulators <i>(Nash Equilibrium — now the stable AND best collective outcome)</i>	A bears moderation costs but gains regulatory favour. B faces penalties and reputational damage for non-compliance <i>(B is punished — governance protects A's investment)</i>
Platform A: Lax	A faces penalties and reputational damage for non-compliance. B bears moderation costs but gains regulatory favour <i>(A is punished — governance protects B's investment)</i>	Both face regulatory sanctions, fines, and reputational collapse. No trust from users, advertisers, or regulators <i>(Worst outcome — governance makes mutual defection catastrophic)</i>

The shift: Governance changes the payoff structure so that moderation is now the dominant strategy. Both platforms choosing lax — previously the Nash Equilibrium — is now the worst possible outcome. The game has been redesigned.

INSTRUCTOR NOTES:

Explain that governance does not magically make actors ethical. It changes incentives and constraints. Lax behavior becomes more costly through penalties, audits, reputational damage, or legal exposure. Responsible moderation becomes more valuable through trust, legitimacy, and compliance.

SPEAKING NOTES:

"The lesson is not that regulation solves everything. The lesson is that changing rules, evidence, costs, and authority changes the strategic structure."

Decision Trees

Payoff matrices model simultaneous choices.

Decision trees model sequential choices.

Useful for pathways such as:

User posts → platform flags → user appeals → moderator reviews → regulator responds

INSTRUCTOR NOTES:

Introduce decision trees briefly as a second technique, but do not develop it fully in this deck unless time allows.

Connect decision trees to BBS pathways and temporal sequence. This provides a bridge from game theory back into mediation pathway analysis.

SPEAKING NOTES:

"We will mostly use the payoff matrix today, but decision trees are useful when actions unfold in sequence across a mediation pathway."

Course Framework Connection

Game theory describes strategic structure.

BBS describes mediation pathways.

EDOCA diagnoses:

- Effort
- Distortion
- Observability
- Control
- Authority



INSTRUCTOR NOTES:

Make the relationship among frameworks explicit.

Game theory does not replace BBS or EDOCA. It gives a structure that BBS and EDOCA can analyze.

SPEAKING NOTES:

"Game theory helps us describe the situation. BBS and EDOCA help us ask where the signals move, what gets distorted, who can observe, who can control, and who has authority."

Earlier Bridge: Games and Observability

Earlier examples:

- Chess
- Stratego
- Poker
- Dice
- Austerlitz

Each has a different information structure.



INSTRUCTOR NOTES:

Reconnect to the earlier lecture where these examples were used to discuss observability, hidden information, capabilities, and strategic uncertainty. This avoids making Week 12 feel disconnected.

SPEAKING NOTES:

"We have already started this discussion. We used games to ask: what is visible, what is hidden, what is revealed only through interaction, and what is not yet known?"

Information Structures Revisited

Chess

Open state + hidden strategy

Poker

Hidden state + permitted deception

Austerlitz

Manufactured opacity

Stratego

Visible position + hidden capability

Dice

Future state not yet determined

Visibility is not understanding. Opacity can be manufactured.

INSTRUCTOR NOTES:

This consolidated slide covers all five information-structure examples. Chess has open board state but hidden strategy. Stratego has visible positions but hidden capabilities. Poker has hidden state with permitted deception (bluffing is part of the rules). Dice represent information that does not yet exist. Austerlitz shows manufactured opacity — actors can manipulate what others believe the information structure to be.

Use this slide to distinguish visible state, hidden strategy, hidden capability, permitted deception, future uncertainty, and manufactured opacity. The ethical lesson is that opacity can be produced strategically.

SPEAKING NOTES:

"Chess shows that full visibility does not reveal intent. Stratego hides capability behind visible position. In Poker, bluffing is part of the rules. In dice, the outcome has not yet come into existence. And at Austerlitz, opacity was manufactured deliberately. Visibility is not understanding."

What Games Help Us See

Information may be:

- visible
- hidden
- partially visible
- revealed through interaction
- delayed
- random
- strategically concealed

INSTRUCTOR NOTES:

This is a synthesis slide.

Connect directly to EDOCA observability and BBS mediation.

SPEAKING NOTES:

"Observability is not binary. It varies by agent, by signal, by time, and by pathway."

Bounded Games vs Real Systems

Games are bounded.

Real systems may not be.

In real systems:

- rules change
- actors enter from outside
- enforcement is uneven
- signals cross platforms
- authority is distributed

INSTRUCTOR NOTES:

Important transition.

Do not let students treat games as perfect analogies. Games clarify structure; real sociotechnical systems exceed game boundaries.

SPEAKING NOTES:

"A board game is finite. A data system is embedded in institutions, platforms, markets, laws, users, and bad actors."

Internal Game and Meta-Game

Internal game

Official rules, moves, and outcomes

Meta-game

The larger system of enforcement, cheating, incentives, governance, and external pressure

INSTRUCTOR NOTES:

Define the nested-game idea clearly.

Use tournament chess as an example if useful: board rules are internal; clocks, ratings, referees, communication restrictions, and anti-cheating systems are meta-game structures.

SPEAKING NOTES:

"Cheating often does not occur as a legal move inside the game. It enters through the larger system around the game."

Cheating and System Design

Cheating often exploits:

- weak observability
- weak enforcement
- hidden channels
- uneven authority
- external resources
- unclear rules



INSTRUCTOR NOTES:

Frame cheating as a design problem, not only an individual moral failure.
Connect to EDOCA: observability, control, authority, effort, distortion.

SPEAKING NOTES:

"A system can be designed to make cheating easier or harder. That is an ethical design question."

Bad Information in Game Terms

Bad information can be a signal, a strategy, a distortion, a rule exploit, cheating, or a meta-game intervention.

Misinformation — false or misleading, not necessarily intentional

Disinformation — deliberate deception or manipulation

Malinformation — real information used harmfully or out of context

The same signal can have different ethical meanings depending on who sends it, why, and how the system amplifies it.

INSTRUCTOR NOTES:

Connect Week 11 taxonomy to Week 12 game structure. Bad information is not one thing. Its ethical meaning depends on agency, rules, pathway, and context. This is a recall slide — students encountered the misinformation/disinformation/malinformation taxonomy in Week 11. Use it to prepare the COVID case construction.

The slide now combines strategic roles (signal, strategy, distortion, rule exploit, cheating, meta-game) with the formal taxonomy. Emphasize that the same signal can have different ethical meanings depending on who sends it, why, and how the system amplifies it.

SPEAKING NOTES:

"A misleading claim may be a signal. A coordinated campaign may be a strategy. Bot impersonation may be cheating. The same signal can have different ethical meanings depending on who sends it, why, and how the system amplifies it."

Shared Case

COVID-19 Misinformation Moderation

A platform receives rapidly spreading posts during a public health crisis.

Some are accurate.

Some are uncertain.

Some are misleading.

Some are intentionally manipulative.



INSTRUCTOR NOTES:

Anchor the shared exercise. This continues directly from Week 11.

Do not invite broad COVID policy debate. Keep focus on system specification.

SPEAKING NOTES:

"We are using COVID misinformation because the class already has a shared reference point. Our goal is to model the system, not debate pandemic policy."

Moderation Has Costs

Over-removal may suppress legitimate speech.

Under-removal may amplify harm.

Transparency may help users.

Transparency may help bad actors adapt.

Delay may increase damage.

INSTRUCTOR NOTES:

This slide sets up ethical instability.

Emphasize that every action has tradeoffs. The point is to specify those tradeoffs structurally.

SPEAKING NOTES:

"In this kind of system, doing nothing has consequences. Acting also has consequences."

Class Activity: Build the Game

Shared construction — one team, one case

We will now apply the specification method to the COVID-19 misinformation moderation case — together, as a class.

How this works:

- Work as one design team — not separate groups
- Use the specification structure to define agents, goals, signals, rules, and failure modes
- Build the specification on the board as we go

Goal: Produce a partial specification, not a finished game. The incompleteness is not failure — unresolved questions show where ethical analysis must continue.

THIS IS WHERE WE GOT TO IN CLASS

Agents in the COVID Misinformation Game

Who can act within or upon this system?

- **Platform governance team** — sets moderation policy, deploys tools, responds to pressure
- **Public health institution** — issues guidance, requests enforcement, has limited direct control
- **Ordinary user / community member** — shares, consumes, reports, amplifies
- **Bad actor / misinformation producer** — creates, coordinates, exploits platform rules
- **Regulator / professional oversight body** — sets rules, investigates, sanctions (slow authority actor)
- *Optional: Recommender algorithm* — amplifies, filters, ranks content

Is the recommender an agent or a mediation structure? If the class disagrees, preserve the disagreement — the question itself is analytically useful.

Shared Specification Method

Use this structure:

- | | |
|----------------------------------|---|
| 1. System goal | 6. Information and observability |
| 2. Agents / players | 7. Action frequency, control, authority |
| 3. Goals and victory conditions | 8. Ethical instability |
| 4. Signals and signal powers | 9. Redesign or intervention |
| 5. Rules, constraints, meta-game | 10. Unresolved questions |

INSTRUCTOR NOTES:

This is the activity anchor slide. Keep it visible while beginning the shared construction. The deck provides the conceptual path; the activity guide provides the working scaffold. Students should understand that the detailed board template, prompts, and workflow live in the separate activity guide, not in this slide deck.

Detailed prompts belong in the separate activity guide.

SPEAKING NOTES:

"This is the reusable structure. You will use this today on the shared case, and then again for your final case specification."

Agents and Victory Conditions

Ask:

Who can act?

What does each agent want?

What counts as success?

Where do victory conditions conflict?

INSTRUCTOR NOTES:

This condenses specification steps 2 and 3.

Possible agents: platform governance team, public health institution, ordinary user, misinformation producer, regulator, recommender system.

Preserve ambiguity around whether the recommender is an agent or a mediation structure.

SPEAKING NOTES:

"Local victory can produce system-level harm. That is why the distinction matters."

Signals and Observability

Ask:

What moves through the system?

Who can generate, amplify, verify, distort, suppress, or authorize signals?

What is visible, hidden, delayed, or distorted?

INSTRUCTOR NOTES:

This condenses specification steps 4 and 6.

Signals may include posts, labels, rankings, recommendations, warnings, removals, appeals, policy statements, and public-health guidance.

SPEAKING NOTES:

"Signals are not just content. A warning label is a signal. A ranking decision is a signal. A removal is a signal."

Rules, Meta-Game, Authority

Ask:

What actions are allowed, blocked, delayed, rewarded, or punished?

Who acts quickly?

Who can change the rules?

What larger system surrounds the internal game?

INSTRUCTOR NOTES:

This condenses specification steps 5 and 7.

Examples: platform policy, legal risk, advertiser pressure, public criticism, regulatory pressure, cross-platform coordination, bot networks, media coverage.

SPEAKING NOTES:

"Not all players move at the same speed. Some move once. Some move thousands of times. Some do not move often, but can change the rules."

Ethical Instability and Redesign

Ask:

Where can local success produce wider harm?

What could change?

What new problem would the change create?

What remains unresolved?

INSTRUCTOR NOTES:

This condenses specification steps 8, 9, and 10.

Push students to define ethical instability structurally, not emotionally.

Examples: engagement increases confusion; transparency enables evasion; removal reduces harm but damages trust; delay protects accuracy but permits spread.

SPEAKING NOTES:

"A good redesign changes the game. Ask not only what it fixes, but what new behavior it creates."

Board Output

By the end, we should have:

- system goal
- agents
- victory conditions
- signal types
- fast and slow actors
- meta-game vulnerability
- ethical instability
- unresolved questions

INSTRUCTOR NOTES:

Use this slide to close the shared construction.

It makes the activity concrete and prevents over-expansion.

SPEAKING NOTES:

"We are not building a full game. We are building enough structure to carry into the final case analysis."

Reset: Where Are We Now?

Your final case work begins now.

Identify:

- one concept you understand
- one concept still unclear
- one Lite Case you may select
- one source gap

INSTRUCTOR NOTES:

The reset happens after the method is demonstrated, not before. Students now understand what kind of structure their final case needs.

Use this as a brief check-in.

SPEAKING NOTES:

"Now that we have modeled a shared case, we can ask where each of you stands in relation to your own final case."

Lite Case Study Selection

Select from the provided Lite Case Study packet.

Do not use the full case studies as shortcuts.

The Lite cases provide structure, not answers.

INSTRUCTOR NOTES:

Be explicit. Students are selecting from the prepared Lite packet, not the entire inventory.

The full case studies are instructor-facing or optional references because they already answer too many of the questions students must work through.

SPEAKING NOTES:

"The Lite case gives you enough to begin. Your job is to perform the analysis."

Zotero as Evidence Infrastructure

Zotero is not only citation management.

It is ethical evidence infrastructure.

Poor sources weaken ethical claims.

Missing sources create distortion.

INSTRUCTOR NOTES:

This is a key conceptual slide.

Frame evidence as part of ethical responsibility. Ethical claims depend on trustworthy information.

SPEAKING NOTES:

"If your evidence structure is weak, your ethical claim is weak. If your sources are missing, your analysis is distorted."

Source Ecology

Your final analysis should include, where possible:

- factual system context
- ethical or social consequences
- governance or professional standards
- technical or institutional evidence
- a source that challenges your position

INSTRUCTOR NOTES:

Students should review their Zotero folders against this checklist.

Do not demand impossible source coverage for every case, but push beyond journalism-only evidence.

SPEAKING NOTES:

"A final ethical analysis should not rely only on platform rhetoric or a single news story."

Assignment 12: Final Case Study Specification Draft

Purpose: Define the structure before making the argument.

Use the specification method:

- selected Lite case
- system goal
- agents and victory conditions
- signals and rules / meta-game
- observability and EDOCA diagnosis
- ethical instability
- evidence plan and unresolved question

INSTRUCTOR NOTES:

Introduce the assignment briefly. The full assignment should be distributed separately as an assignment guide. Emphasize that this is not the final paper — it is preparation for the final paper. The skeleton (selected Lite case, system goal, agents and victory conditions, signals and rules, observability and EDOCA diagnosis, ethical instability, evidence plan) gives students the structure without turning the slide deck into the full assignment sheet.

Detailed word count, GitHub path, and requirements should live in the separate assignment guide.

SPEAKING NOTES:

"You are not writing the full argument yet. You are building the structure that makes a responsible argument possible. If you can complete these headings clearly, you will have the foundation for the final paper."

Reflection: Personal and Professional Ethics

Private reflection:

What ethical instability in your case seems hardest to resolve?

What does it reveal about your own ethical commitments?

What would a responsible professional be obligated to do?



INSTRUCTOR NOTES:

This replaces the opening provocation. It looks forward to the personal and professional ethical framework assignment.

Students should write privately.

SPEAKING NOTES:

"Your final case is not only about an external system. It is also evidence of how you reason as a future professional."

Final Synthesis

Ethical systems involve:

- agents
- goals
- signals
- rules
- information limits
- incentives
- authority
- instability
- evidence

INSTRUCTOR NOTES:

Summarize the session in structural terms.

Do not introduce new content.

SPEAKING NOTES:

"Ethical analysis becomes stronger when we can specify the system before judging it."

Forward Link

Next steps:

- Case specification
- Evidence development
- Final paper
- Personal and professional ethical framework



INSTRUCTOR NOTES:

End by connecting the remaining course arc.

The final weeks move from case structure to argument and then to personal/professional ethical articulation.

SPEAKING NOTES:

"The case study gives us the external system. The ethical framework asks how you would reason and act within such a system."